

Technical Document #243: Database Systems - Comprehensive Overview

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Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Key Takeaways:

- Database indexing improves query performance by creating data structures that allow faster data retrieval.
- Database monitoring tracks performance metrics and identifies bottlenecks.
- Database connection pooling reuses database connections to reduce overhead.